

CHAPTER – 2

Purplle Online Beauty Shopping Management System

2.0 Introduction

The entities required for the **Purplle Online Beauty Shopping Management System** are identified based on the business requirements of a beauty marketplace platform. In addition to core e-commerce entities (Customer, Product, Orders, Payment, Delivery), this design reflects Purplle's real-world model as a multi-seller marketplace: independent sellers list products, products come in multiple variants (shade, size, or pack type), customers can maintain a wishlist alongside their cart, and orders can apply coupon-based discounts. Each entity contains attributes along with primary keys, foreign keys, and unique keys to organize and manage this data efficiently while maintaining data integrity and minimal redundancy.

2.1 Customer

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique customer ID
First_Name	VARCHAR(50)	-	Customer first name
Last_Name	VARCHAR(50)	-	Customer last name
Gender	VARCHAR(10)	-	Customer gender
Email	VARCHAR(100)	Unique Key	Email address, must be unique per customer
Phone	VARCHAR(15)	-	Contact number
Address	TEXT	-	Customer address
City	VARCHAR(50)	-	City
State	VARCHAR(50)	-	State
Pincode	INT	-	Postal code

2.2 Category

Attribute	Data Type	Key	Description
Category_ID	INT	PK	Unique category ID
Category_Name	VARCHAR(50)	-	Beauty category name
Description	TEXT	-	Category description

2.3 Brand

Attribute	Data Type	Key	Description
Brand_ID	INT	PK	Unique brand ID
Brand_Name	VARCHAR(50)	Unique Key	Brand name, must be unique

2.4 Seller (New)

Attribute	Data Type	Key	Description
Seller_ID	INT	PK	Unique seller/vendor ID
Seller_Name	VARCHAR(100)	-	Business or store name
Email	VARCHAR(100)	Unique Key	Seller contact email
Phone	VARCHAR(15)	-	Seller contact number
GSTIN	VARCHAR(20)	Unique Key	Tax registration number
Warehouse_Address	TEXT	-	Dispatch location
Rating	DECIMAL(3,2)	-	Average seller rating

2.5 Product

Attribute	Data Type	Key	Description
Product_ID	INT	PK	Unique product ID
Product_Name	VARCHAR(100)	-	Product name
Category_ID	INT	FK	References Category
Brand_ID	INT	FK	References Brand
Seller_ID	INT	FK	References Seller who lists this product
Admin_ID	INT	FK	References Admin who approves/manages listing
Description	TEXT	-	Product description
Expiry_Date	DATE	-	Product expiry date

2.6 Product_Variant (New)

Attribute	Data Type	Key	Description
Variant_ID	INT	PK	Unique variant ID
Product_ID	INT	FK	References Product
Shade_Or_Type	VARCHAR(50)	-	e.g. Shade, size, or pack type
Price	DECIMAL(10,2)	-	Price for this variant
Weight	VARCHAR(20)	-	Product weight/volume
Stock	INT	-	Available quantity for this variant

2.7 Cart

Attribute	Data Type	Key	Description
Cart_ID	INT	PK	Unique cart ID
Customer_ID	INT	FK, Unique Key	References Customer; unique enforces 1 cart per customer

2.8 Cart_Item

Attribute	Data Type	Key	Description
CartItem_ID	INT	PK	Unique cart item ID
Cart_ID	INT	FK	References Cart
Variant_ID	INT	FK	References Product_Variant
Quantity	INT	-	Quantity selected

2.9 Wishlist (New)

Attribute	Data Type	Key	Description
Wishlist_ID	INT	PK	Unique wishlist ID
Customer_ID	INT	FK, Unique Key	References Customer; unique enforces 1 wishlist per customer

2.10 Wishlist_Item (New)

Attribute	Data Type	Key	Description
WishlistItem_ID	INT	PK	Unique wishlist item ID
Wishlist_ID	INT	FK	References Wishlist
Product_ID	INT	FK	References Product
Date_Added	DATE	-	Date product was saved

2.11 Coupon (New)

Attribute	Data Type	Key	Description
Coupon_ID	INT	PK	Unique coupon ID
Coupon_Code	VARCHAR(30)	Unique Key	Code entered at checkout
Discount_Type	VARCHAR(20)	-	Flat or Percentage
Discount_Value	DECIMAL(10,2)	-	Discount amount or percent
Valid_From	DATE	-	Start of validity
Valid_Till	DATE	-	End of validity
Min_Order_Amount	DECIMAL(10,2)	-	Minimum cart value to apply

2.12 Orders

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Coupon_ID	INT	FK	References Coupon, if applied
Order_Date	DATE	-	Date of order
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Pending/Shipped/Delivered/Returned

2.13 Order_Details

Attribute	Data Type	Key	Description
Order_Detail_ID	INT	PK	Unique order detail ID
Order_ID	INT	FK	References Orders
Variant_ID	INT	FK	References Product_Variant
Seller_ID	INT	FK	References Seller fulfilling this line item
Quantity	INT	-	Ordered quantity
Unit_Price	DECIMAL(10,2)	-	Price per unit at time of order

2.14 Payment

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK, Unique Key	References Orders; unique enforces one payment per order
Payment_Method	VARCHAR(30)	-	UPI/Card/COD/Wallet
Payment_Status	VARCHAR(20)	-	Payment status
Payment_Date	DATE	-	Payment date
Amount	DECIMAL(10,2)	-	Paid amount

2.15 Delivery

Attribute	Data Type	Key	Description
Delivery_ID	INT	PK	Unique delivery ID
Order_ID	INT	FK, Unique Key	References Orders; unique enforces one delivery per order
Delivery_Address	TEXT	-	Delivery address
Delivery_Date	DATE	-	Delivery date
Delivery_Status	VARCHAR(30)	-	Order delivery status

2.16 Review

Attribute	Data Type	Key	Description
Review_ID	INT	PK	Unique review ID
Customer_ID	INT	FK	References Customer
Product_ID	INT	FK	References Product
Rating	INT	Check (1-5)	Star rating given
Review_Text	TEXT	-	Customer's written feedback
Review_Date	DATE	-	Date of review

2.17 Admin

Attribute	Data Type	Key	Description
Admin_ID	INT	PK	Unique admin ID
Admin_Name	VARCHAR(100)	-	Administrator name
Email	VARCHAR(100)	Unique Key	Admin email, must be unique
Phone	VARCHAR(15)	-	Contact number
Username	VARCHAR(50)	Unique Key	Login username, must be unique
Password	VARCHAR(100)	-	Login password

2.18 Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Orders	1 : M
Customer	Owns	Cart	1 : 1
Customer	Owns	Wishlist	1 : 1
Customer	Writes	Review	1 : M
Cart	Contains	Cart_Item	1 : M
Cart_Item	Refers To	Product_Variant	M : 1
Wishlist	Contains	Wishlist_Item	1 : M
Wishlist_Item	Refers To	Product	M : 1
Category	Contains	Product	1 : M
Brand	Manufactures	Product	1 : M
Seller	Lists	Product	1 : M
Product	Has	Product_Variant	1 : M
Product	Receives	Review	1 : M
Orders	Contains	Order_Details	1 : M
Product_Variant	Appears In	Order_Details	1 : M
Seller	Fulfil	Order_Details	1 : M
Coupon	Applied To	Orders	1 : M
Orders	Has	Payment	1 : 1
Orders	Has	Delivery	1 : 1
Admin	Manages	Product	1 : M